



ALHUSNA HANIFAH

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SUMMARY

Fresh Graduate in Informatics from Syiah Kuala University, graduated Cum Laude with a GPA of 3.90/4.00. Experienced in applying machine learning and deep learning to develop AI-powered software and full-stack web applications. Contributed to technology-driven solutions through undergraduate research, national innovation competitions, and collaborative software development projects. Awarded the Bronze Medal at the 2025 National Student Digital Innovation Competition (LIDM) and recognized as a Finalist in LIDM 2024. A proactive team player with leadership experience, strong analytical thinking, and problem-solving skills, passionate about integrating artificial intelligence with modern software engineering practices to build intelligent, reliable, and user-centered software solutions.

EXPERIENCE

Laboratory Assistant, Department of Informatics

Aceh, Feb 2024 - DeC 2025

University of Syiah Kuala

- Mentored undergraduate students in laboratory sessions for Programming, Data Mining, Data Structures and Algorithms, Theory of Languages and Automata, and Data Science courses.
- Guided students in understanding algorithms, programming concepts, and data analysis through hands-on laboratory instruction, technical discussions, and code debugging.
- Developed laboratory materials, programming exercises, and assessment components to support effective teaching and learning.
- Collaborated with faculty members and fellow teaching assistants to ensure the smooth and effective delivery of laboratory sessions.

PROJECTS

Circuit Virtual Interactive Application CRVIA | Competition Project
(LIDM 2025 - Bronze Medalist)

Surabaya, 2025

Technologies: Next.js, TypeScript, React, Tailwind CSS, MediaPipe, TensorFlow.js, Supabase

- Collaborated in the design and development of a web-based physics learning platform to help high school students understand electrical circuits through interactive simulations.
- Developed the Interactive Learning Materials feature by implementing both frontend and backend components, delivering an engaging, responsive, and efficient user experience.
- Integrated MediaPipe and TensorFlow.js to enable virtual laboratory activities powered by computer vision.
- Contributed to requirements analysis, system testing, technical documentation, and the presentation of the solution at a national innovation competition.
- Participated in the copyright registration process with the Directorate General of Intellectual Property (DGIP) of Indonesia.

Automatic Descriptive Text Generation on Radiological Images | Undergraduate Thesis

Aceh, 2025-2026

Technologies: Python, PyTorch, Hugging Face Transformers, FastAPI, MedCLIP, BioBART

- Developed an AI-powered image captioning system to automatically generate diagnostic descriptions from radiology images using a vision-language approach.
- Integrated MedCLIP as the visual encoder and BioBART as the language model through a custom mapping network to bridge visual representations and natural language generation.
- Built an end-to-end inference pipeline using PyTorch and deployed the model with FastAPI, exposing it as a RESTful API.
- Performed data preprocessing, model experimentation, performance evaluation, and inference optimization throughout the research lifecycle.

AR Molecular Explorer (LIDM 2024 - National Finalist)| Competition Project Bogor, 2024
Technologies: Unity, Augmented Reality (AR)

- Led the development of AR Molecular Explorer, an Augmented Reality-based educational application designed to help high school students visualize molecular geometry through interactive learning experiences.
- Coordinated requirements analysis, solution design, application development, testing, and task allocation to ensure effective collaboration and timely project delivery.
- Contributed to the implementation of core application features and the preparation of presentation materials for a national innovation competition.
- Presented the solution to judging panels at both university and national levels, demonstrating the application's technical implementation, educational value, and potential impact on science learning.

EDUCATION

Bachelor of Informatics | Syiah Kuala University • 2026 • 3.90 **Banda Aceh, Aceh 2022-2026**

- IPK: 3,90/4,00
- Undergraduate Thesis: Automatic Descriptive Text Generation on Radiological Images Using MedCLIP and BioBART Model
- Graduated Cum Laude

SKILL

Technical Skills

- Programming Languages: Python, PHP, JavaScript, Java
- AI & Machine Learning: TensorFlow, PyTorch, Scikit-learn, OpenCV, Transformers, MediaPipe
- Frameworks: Next.js, React, Django, AdonisJS, Vue.js, Laravel
- Database: MongoDB, MySQL, Supabase
- Tools: Git, GitHub, Docker, Figma, FastAPI

Professional Skills

• Leadership • Problem Solving • Team Collaboration • Communication • Analytical Thinking • Adaptability

INVOLVEMENT

Vice Head | Department of Student Welfare Management (PKM) **Aceh, 2023 -2024**
Informatics Student Association

- Assisted in planning, coordinating, and implementing programs aimed at enhancing student welfare, personal development, and engagement.
- Collaborated with department officers and student representatives to ensure the successful execution of organizational programs and activities.
- Contributed to departmental decision-making, program evaluation, and administrative operations to support effective organizational management.

ACHIEVEMENT

3rd Place (Bronze Medalist) at National Digital Innovation Competition | Kemdiktisaintek • 2025

- Awarded the Bronze Medal for an innovative educational technology solution.

Finalist at National Digital Innovation Competition | Kemdiktisaintek • 2024

- Recognized as a National Finalist for an innovative educational technology solution.

SOCIAL MEDIA

Instagram: alhusna_hanifah